

# A deformability threshold for stiffness-based sorting in deterministic-lateral-displacement microfluidic devices: a 2D lattice-Boltzmann immersed-boundary parameter sweep

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## Abstract

Deterministic lateral displacement (DLD) is the workhorse passive-sorting technology in microfluidic cell-separation chips, but the classical Huang–Holm critical-bin theory was derived for rigid spheres. Recent experimental and computational work has shown that deformable particles can be sorted by stiffness at identical particle size, opening the door to label-free, mechanophenotyping-based separation of red blood cells, circulating tumour cells and synthetic vesicle carriers. The quantitative conditions under which stiffness alone is sufficient to drive sorting, however, remain unclear. Here we use *a custom-built 2D lattice-Boltzmann immersed-boundary code with Skalak-membrane capsules* (developed in-house; equations and full discretisation given in Section 2) to perform a systematic upper-triangle parameter sweep over 15  $(G_s^{\text{soft}}, G_s^{\text{stiff}})$  pairs spanning a  $27\times$  stiffness contrast, with all other membrane parameters matched so the diagonal  $G_s^{\text{soft}} = G_s^{\text{stiff}}$  is a strict no-contrast control. The diagonal-subtracted DLD displacement-angle separation,  $\Delta\theta_{\text{corr}}$ , isolates the pure deformability contribution from the finite- $N$  statistical baseline. We identify a clear *deformability threshold*: stiffness-based sorting emerges only when the soft-species Capillary number exceeds  $\text{Ca}_{\text{soft}} \approx 0.2$ , below which both species traverse the array as effectively rigid spheres and the geometric DLD selection rule applies to both regardless of contrast. Above the threshold,  $|\Delta\theta_{\text{corr}}|$  grows with the stiffness ratio and saturates near a ratio of  $\sim 13$ . This practical design rule — operate at  $\text{Ca}_{\text{soft}} > 0.2$  for deformability sorting — directly informs chip parameter selection (pillar gap, throughput, target cell type).

**Keywords:** deterministic lateral displacement · capillary number · capsule deformability · lattice Boltzmann method · immersed boundary method · cell sorting · microfluidics · mechanophenotyping

## 1 Introduction

Deterministic lateral displacement (DLD) [4, 8] sorts suspended particles by size using a parallelogram lattice of micropillars whose row-to-row lateral shift  $\delta$  defines a *critical bin* diameter  $D_c$ . Particles larger than  $D_c$  are “bumped” laterally one row per column — the so-called displacement mode — while smaller particles execute a zigzag trajectory and recover their initial streamline after a full lattice period. The original derivation by Davis and co-workers [4, 9] assumed rigid spherical particles, and Holm and co-workers later refined the critical-size formula to

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$D_c \approx 1.4 G \sqrt{\delta/\Delta x}$  for hard spheres in symmetric arrays [6], where  $G$  is the lateral pillar gap and  $\Delta x$  the column spacing. This rigid-sphere theory has been verified experimentally [16] and is now the working basis of clinical DLD chips for circulating-tumour-cell enrichment and red-blood-cell fractionation.

A growing body of work shows that the rigid-sphere assumption breaks down for biological cells, whose deformability fundamentally alters their interaction with the obstacle array [1, 18, 22]. Holmes et al. [7] demonstrated experimentally that softer red blood cells (treated to alter their elasticity) migrate at a smaller effective displacement angle than stiffer ones *at identical size*, opening DLD to mechanophenotyping. On the computational side, Krüger et al. [12] used a 3D lattice-Boltzmann immersed-boundary finite-element framework to quantify deformability-driven critical-bin shifts in DLD; Ye et al. [21] showed by direct numerical simulation that the critical separation diameter is shifted by 10–40 % depending on the Capillary number; Kabacaoglu and Biros [10] extended the analysis to “deep” DLD geometries where the third dimension is exploited. Vahidkhah and Bagchi [20] examined capsule shape effects in 3D suspensions, and D’Avino and Maffettone [3] reviewed the broader class of softness-driven migration in confined microchannel flows.

What remains poorly mapped is the *threshold* behaviour: at which value of the soft-species Capillary number,  $Ca_{\text{soft}}$ , does deformability begin to contribute appreciably to sorting? Below the threshold, both soft and stiff particles should behave as effectively rigid and any stiffness contrast should be irrelevant. Above it, a chip operating in deformability-sorting mode should display a monotonically increasing displacement-angle separation with stiffness ratio. The threshold itself is the key engineering design rule for any DLD device targeting mechanophenotyping — it sets the minimum required flow speed (and hence chip throughput) for a given target cell type, or conversely the maximum allowed stiffness for which sorting is feasible at a given throughput.

This study reports a systematic 2D lattice-Boltzmann immersed-boundary parameter sweep over a five-by-five grid of soft and stiff shear-elastic moduli, with all other capsule mechanical parameters held identical so that the diagonal of the grid is a strict no-contrast control. All simulations were performed with a custom code developed by the author & its team at the Department of Natural Sciences, the Begum Nusrat Bhutto Women University, Sukkur ([www.bnbwu.edu.pk](http://www.bnbwu.edu.pk)); the discretisation, constitutive equations, and numerical schemes are documented in full in Section 2. We introduce a diagonal-subtracted displacement-angle separation,  $\Delta\theta_{\text{corr}}$ , that isolates the pure deformability contribution from the finite- $N$  noise inherent in any particle-resolved suspension simulation, and we identify a deformability threshold at  $Ca_{\text{soft}} \approx 0.2$  above which sorting emerges.

The remainder of this paper is organised as follows. Section 2 describes the LBM-IBM engine, the capsule membrane model, the DLD geometry, and the sweep design including the randomised-order singleton seeding protocol we developed to remove two-pass RSA bias. Section 3 reports the diagonal noise-floor characterisation, the corrected angular separation, and the Capillary-number collapse that identifies the threshold. Section 4 compares the threshold to published 3D data and discusses limitations. Section 5 summarises the design rule and outlines future work.

## 2 Methods

This section documents the full numerical formulation of SOFTFLOW, the author’s in-house 2D lattice-Boltzmann immersed-boundary code, in sufficient detail that any independent implementation can reproduce the present results without reference to a particular software release. Section 2.1 gives the fluid solver, Section 2.2 the boundary conditions, Section 2.3 the fluid–solid coupling, and Section 2.4 the capsule constitutive model. The DLD geometry and sweep design are in Sections 2.5–2.9.

Throughout the paper all quantities are reported in lattice units (LU): the lattice spacing  $\Delta x$ , the timestep  $\Delta t$  and the reference density  $\rho_0$  are each set to unity. Conversion to physical units is treated in Section 4.1.

## 2.1 Fluid solver: D2Q9 lattice Boltzmann equation

The fluid is solved on a regular two-dimensional Cartesian grid by the lattice Boltzmann method [13] with the D2Q9 velocity set

$$\mathbf{e}_i = \begin{cases} (0, 0) & i = 0, \\ (\pm 1, 0), (0, \pm 1) & i = 1, \dots, 4, \\ (\pm 1, \pm 1) & i = 5, \dots, 8, \end{cases} \quad (1)$$

with weights  $w_0 = 4/9$ ,  $w_{1-4} = 1/9$ ,  $w_{5-8} = 1/36$  and lattice speed of sound  $c_s^2 = 1/3$ .

Let  $f_i(\mathbf{x}, t)$  be the discrete density distribution in direction  $i$  at lattice site  $\mathbf{x}$  and time  $t$ . Macroscopic moments are

$$\rho(\mathbf{x}, t) = \sum_{i=0}^8 f_i, \quad \rho \mathbf{u}(\mathbf{x}, t) = \sum_{i=0}^8 f_i \mathbf{e}_i + \frac{1}{2} \mathbf{F} \Delta t, \quad (2)$$

where the half-force correction in the momentum is required for consistency with the second-order Guo forcing scheme used below.

The equilibrium distribution is the Hermite-truncated Maxwellian

$$f_i^{\text{eq}}(\rho, \mathbf{u}) = w_i \rho \left[ 1 + \frac{\mathbf{e}_i \mathbf{u}}{c_s^2} + \frac{(\mathbf{e}_i \mathbf{u})^2}{2c_s^4} - \frac{\mathbf{u} \mathbf{u}}{2c_s^2} \right]. \quad (3)$$

The collide-stream cycle is

$$f_i^*(\mathbf{x}, t) = f_i(\mathbf{x}, t) - \frac{1}{\tau} [f_i(\mathbf{x}, t) - f_i^{\text{eq}}(\mathbf{x}, t)] + F_i(\mathbf{x}, t) \Delta t, \quad (4)$$

$$f_i(\mathbf{x} + \mathbf{e}_i \Delta t, t + \Delta t) = f_i^*(\mathbf{x}, t), \quad (5)$$

with relaxation time  $\tau$  related to the kinematic viscosity by  $\nu = c_s^2(\tau - \frac{1}{2}) \Delta t = (\tau - \frac{1}{2})/3$ . We use  $\tau = 0.70$ , giving  $\nu = 0.0667$  LU throughout the sweep. The bulk force  $\mathbf{F}$  enters the collision step through Guo et al. second-order forcing,

$$F_i = \left(1 - \frac{1}{2\tau}\right) w_i \left[ \frac{\mathbf{e}_i - \mathbf{u}}{c_s^2} + \frac{(\mathbf{e}_i \mathbf{u}) \mathbf{e}_i}{c_s^4} \right] \cdot \mathbf{F}. \quad (6)$$

The driving body force is  $\mathbf{F} = (F_x, 0)$  with  $F_x = 3 \times 10^{-5}$  LU, producing an empty-channel Poiseuille centre-line velocity  $U_c = F_x H^2 / (8\nu) \approx 0.035$  LU and a Mach number  $\text{Ma} = U_c / c_s \approx 0.06$ , well within the LBM incompressibility regime.

**Regularised collision.** The standard BGK collision (4) is unstable for  $\tau$  close to  $\frac{1}{2}$  because of unphysical ghost modes carried by the non-hydrodynamic part of  $f_i$ . We adopt the *regularised pre-collision* projection of Latt and Chopard [14]: the non-equilibrium increment  $f_i^{\text{neq}} = f_i - f_i^{\text{eq}}$  is replaced before collision by its projection onto the second-order Hermite subspace,

$$\tilde{f}_i^{\text{neq}} = w_i \frac{1}{2c_s^4} \mathbf{H}_i^{(2)} : \mathbf{\Pi}^{\text{neq}}, \quad \mathbf{\Pi}^{\text{neq}} = \sum_j \mathbf{e}_j \mathbf{e}_j f_j^{\text{neq}}, \quad (7)$$

where  $\mathbf{H}_i^{(2)} = \mathbf{e}_i \mathbf{e}_i - c_s^2 \mathbf{I}$  is the second-order Hermite tensor and “:” denotes the double-contraction. The regularised projection damps the ghost modes and substantially improves stability at low  $\tau$  with no additional kernel cost.

## 2.2 Boundary conditions

Three kinds of boundary appear in our domain.

**Periodic.** At the left and right edges, distributions are streamed periodically.

**No-slip channel walls.** At  $y = 0$  and  $y = N_y$  a halfway bounce-back rule is applied: any post-collision distribution that would stream into the wall is reflected at the half-link,

$$f_{\bar{i}}(\mathbf{x}_b, t + \Delta t) = f_i^*(\mathbf{x}_b, t), \quad (8)$$

where  $\bar{i}$  denotes the velocity opposite to  $i$  and  $\mathbf{x}_b$  the boundary node. This gives the no-slip condition with effective wall position located  $\frac{1}{2}\Delta x$  inside the boundary layer, second-order accurate in  $\Delta x$ .

**Curved pillars: interpolated bounce-back.** For circular pillars whose surfaces cut lattice links at fractional distances  $q \in [0, 1)$  from the fluid node, the halfway rule of Eq. (8) is replaced by the linear interpolated bounce-back of Bouzidi, Firdaouss and Lallemand [2],

$$f_{\bar{i}}(\mathbf{x}_b, t + \Delta t) = \begin{cases} 2q f_i^*(\mathbf{x}_b, t) + (1 - 2q) f_i^*(\mathbf{x}_b - \mathbf{e}_i \Delta t, t) & q < \frac{1}{2}, \\ \frac{1}{2q} f_i^*(\mathbf{x}_b, t) + \frac{2q-1}{2q} f_{\bar{i}}^*(\mathbf{x}_b, t) & q \geq \frac{1}{2}, \end{cases} \quad (9)$$

which preserves second-order spatial accuracy on the curved surface.

### 2.3 Fluid–particle coupling: immersed boundary method

Each capsule membrane is discretised as a closed polygon of  $N_{\text{node}}$  Lagrangian markers  $\mathbf{X}_l$ ,  $l = 1, \dots, N_{\text{node}}$ , connected by springs (Section 2.4). Coupling to the LBM fluid uses the Peskin immersed-boundary method [17].

The membrane force at each marker,  $\mathbf{F}_l$ , is spread to the surrounding Eulerian lattice sites by

$$\mathbf{f}(\mathbf{x}, t) = \sum_{l=1}^{N_{\text{node}}} \mathbf{F}_l(t) \delta_h(\mathbf{x} - \mathbf{X}_l) \Delta s_l, \quad (10)$$

where  $\Delta s_l$  is the discrete arc length associated with marker  $l$  and  $\delta_h$  is the four-point Peskin regularised delta function,

$$\delta_h(\mathbf{r}) = \phi(r_x/\Delta x) \phi(r_y/\Delta x) / \Delta x^2, \quad \phi(r) = \begin{cases} \frac{1}{8} (3 - 2|r| + \sqrt{1 + 4|r| - 4r^2}) & 0 \leq |r| < 1, \\ \frac{1}{8} (5 - 2|r| - \sqrt{-7 + 12|r| - 4r^2}) & 1 \leq |r| < 2, \\ 0 & |r| \geq 2. \end{cases} \quad (11)$$

The spread force  $\mathbf{f}(\mathbf{x}, t)$  enters Eq. (6) as part of the total  $\mathbf{F}$  acting on the fluid.

After the collide-stream step, the new fluid velocity is interpolated back to the markers,

$$\mathbf{U}_l(t + \Delta t) = \sum_{\mathbf{x}} \mathbf{u}(\mathbf{x}, t + \Delta t) \delta_h(\mathbf{x} - \mathbf{X}_l) \Delta x^2, \quad (12)$$

and the markers advect with the local fluid velocity,  $\mathbf{X}_l(t + \Delta t) = \mathbf{X}_l(t) + \mathbf{U}_l \Delta t$ .

**Multi-direct forcing iteration.** A single pass of Eqs. (10)–(12) leaves a residual no-slip violation  $\Delta \mathbf{U}_l = \mathbf{U}_l - \dot{\mathbf{X}}_l$  of order  $0.1 U_c$  at the membrane. We use the multi-direct-forcing correction of Luo et al. [15]: for  $K = 2$  iterations  $k = 1, 2$ ,

$$\mathbf{F}_l^{(k+1)} = \mathbf{F}_l^{(k)} + \rho \frac{\dot{\mathbf{X}}_l - \mathbf{U}_l^{(k)}}{\Delta t}, \quad (13)$$

re-spreading via Eq. (10) after each update. Two iterations reduce  $\|\Delta \mathbf{U}_l\|$  by an order of magnitude, which is critical for mass conservation in the densely-packed inlet region.

## 2.4 Capsule membrane model

Each capsule of nominal radius  $R$  is represented by  $N_{\text{node}} = 18$  markers arranged on a closed polygon with node spacing  $\Delta s \approx 2\pi R/N_{\text{node}} \approx 1.05$  LU, consistent with the IBM kernel resolution. Three energetic contributions act on each membrane.

**In-plane shear and area dilation: Skalak constitutive law.** For each membrane edge  $e_l = \mathbf{X}_{l+1} - \mathbf{X}_l$  we compute the local in-plane stretches relative to the equilibrium reference configuration. Approximating the capsule locally as a 1D membrane strip, the Skalak strain-energy density [19] takes the explicit form

$$W_{\text{Skalak}} = \frac{G_s}{4} [(\lambda_1^2 + \lambda_2^2 - 2)^2 + 2(\lambda_1^2 + \lambda_2^2 - 2) - 2(\lambda_1^2 \lambda_2^2 - 1)] + \frac{C}{4} (\lambda_1^2 \lambda_2^2 - 1)^2, \quad (14)$$

where  $\lambda_1 = |\mathbf{e}_l|/|\mathbf{e}_l^{(0)}|$  is the in-plane principal stretch,  $\lambda_2 \approx 1$  for the thin-strip discretisation,  $G_s$  is the surface shear modulus and  $C$  is the area-dilatation modulus. The latter is held at  $C = 10G_s$  throughout the sweep, the conventional value for area-conserving red-blood-cell-like membranes [13]. The node force is obtained by differentiating  $W$  with respect to  $\mathbf{X}_l$ ,  $\mathbf{F}_l^{\text{shear}} = -\partial W_{\text{Skalak}}/\partial \mathbf{X}_l$ , which we evaluate by central finite differences.

**Bending: angular spring potential.** Membrane bending is resolved at each interior node  $l$  via the discrete angle  $\theta_l$  between successive edges  $\mathbf{e}_{l-1}$  and  $\mathbf{e}_l$ , contributing an energy [11]

$$U_{\text{bend}}^{(l)} = \frac{k_{\text{bend}}}{2} (\theta_l - \theta_l^{(0)})^2, \quad (15)$$

with reference angle  $\theta_l^{(0)} = 2\pi/N_{\text{node}}$  for the equilibrium circular shape. The bending force at node  $l$  has components on  $\mathbf{X}_{l-1}$ ,  $\mathbf{X}_l$  and  $\mathbf{X}_{l+1}$  obtained by analytic differentiation.

**Area and perimeter conservation.** Global penalty terms enforce capsule incompressibility against spurious area drift and perimeter creep:

$$U_{\text{area}} = \frac{k_{\text{area}}}{2} (A - A_0)^2/A_0, \quad U_{\text{perim}} = \frac{k_{\text{perimeter}}}{2} (P - P_0)^2/P_0, \quad (16)$$

where  $A$ ,  $P$  are the instantaneous polygon area and perimeter and  $A_0 = \pi R^2$ ,  $P_0 = 2\pi R$  their equilibrium values. The corresponding nodal forces are distributed uniformly over the membrane.

**Matched parameters between species.** For the sweep, the only parameter that distinguishes the two species is the surface shear modulus  $G_s$ . The bending, area- and perimeter-conservation moduli are held identical:  $k_{\text{bend}} = 0.010$ ,  $k_{\text{area}} = 0.75$ ,  $k_{\text{perimeter}} = 0.075$ , all in lattice units. This matched-parameter design is critical for the analysis: the diagonal of the sweep grid ( $G_s^{\text{soft}} = G_s^{\text{stiff}}$ ) is then a *strict no-contrast control* in which the two species are mechanically identical and the displacement-angle separation must vanish in expectation.

**Short-range repulsion and lubrication.** To prevent membrane interpenetration at close encounters between capsules or between capsules and pillars, a Hertzian repulsion is applied when the surface-to-surface gap  $h$  falls below a cutoff  $h_{\text{cut}} = 1.5$  LU,

$$F_{\text{rep}} = k_{\text{rep}} (h_{\text{cut}} - h)^{3/2} \quad \text{for } h < h_{\text{cut}}. \quad (17)$$

A sub-grid lubrication correction is additionally activated [11] whenever capsule-pillar or capsule-capsule gaps fall below the LBM resolution limit, regularised by a minimum-gap cutoff  $h_{\text{min}} = 0.1$  LU to prevent the  $1/h$  singularity.

## 2.5 DLD geometry

The lattice domain is  $N_x \times N_y = 400 \times 80$  in lattice units, periodic in  $x$ . The pillar array consists of  $8 \times 4$  circular obstacles arranged on a parallelogram lattice:

- Pillar radius  $R_p = 4$  LU.
- Column spacing  $\Delta x = 35$  LU.
- Row spacing  $\Delta y = 14$  LU.
- Row-to-row lateral shift  $\delta = 3.5$  LU per column.

The DLD slope is therefore  $\delta/\Delta x = 0.1$  and the predicted critical bin for rigid spheres [6] is  $D_c/G \approx 1.4\sqrt{\delta/\Delta x} \approx 0.44$ , giving  $D_c \approx 12.6$  LU in the gap-width units of our array. Our particle radius is  $R = 3$  LU so that  $2R/D_c \approx 0.48$ , i.e. the particles sit *below* the rigid-sphere critical bin. Any displacement-angle separation we measure between soft and stiff species at identical  $R$  is therefore attributable to deformability alone — both species are sub-critical in the rigid limit.

## 2.6 Parameter sweep grid

The shear modulus is varied over five values,

$$G_s \in \{0.030, 0.100, 0.200, 0.400, 0.800\} \text{ LU}, \quad (18)$$

giving Capillary numbers  $\text{Ca} = \mu\dot{\gamma}R/G_s$  that span more than a decade (Table 1). The grid is upper-triangular plus diagonal,  $G_s^{\text{soft}} \leq G_s^{\text{stiff}}$ , for a total of 15 unique cells. Each cell is one independent simulation.

## 2.7 Particle seeding: randomised-order singleton placement

Naive two-pass random sequential addition (RSA) into a shared inlet region biases the second species systematically: the first pass claims the best positions, the second pass then faces an already-crowded RSA problem near jamming and aborts most placements. In our DLD geometry an inlet of  $42 \times 64$  LU is at the edge of RSA jamming for 40 capsules of  $R = 3$  with min-gap = 1.5; naive two-pass seeding gave a 20/11 species imbalance with  $\Delta\theta_{\text{diag}} \approx 3^\circ$  on the no-contrast diagonal. We adopt instead a *randomised-order singleton seeding* protocol: a list of 20 “soft” + 20 “stiff” labels is constructed, shuffled with the cell’s RNG seed, and capsules are placed one at a time in the shuffled order. The two species see statistically identical placement densities at every step, and after applying a hard cap of 500 placement attempts per singleton, 35 of 40 capsules are reproducibly seeded (typically 17 soft + 18 stiff or vice versa). This 5/40 rejection rate is a stationary property of the RSA problem at our gap setting, not a bias source.

## 2.8 Warm-up protocol

To prevent spurious shocks from the initially overlapping membrane-fluid coupling, each simulation is run for a two-phase warm-up before data collection begins:

1. **500 steps with zero body force:** capsule overlaps relax via mutual repulsion and membrane elasticity.
2. **1500 steps with linearly ramped body force from 0 to  $F_x = 3 \times 10^{-5}$ :** avoids the pressure shock of an instantaneous flow start.

The 2000-step warm-up is then followed by 20 000 production steps, written to disk every 25 steps for trajectory analysis.

## 2.9 Diagnostics: corrected displacement angle

The DLD displacement angle for each species is computed from the mean per-particle displacement between the first and the last trajectory snapshot:

$$\theta_s = \arctan\left(\frac{\langle \Delta y \rangle_s}{\langle \Delta x \rangle_s}\right), \quad s \in \{\text{soft}, \text{stiff}\}, \quad (19)$$

with the periodic- $x$  wrap correction  $\Delta x \rightarrow \Delta x + N_x$  if  $\Delta x < 0$ . The raw displacement-angle separation is  $\Delta\theta_{\text{raw}} = |\theta_{\text{soft}} - \theta_{\text{stiff}}|$ . To isolate the pure deformability-contrast contribution from finite- $N$  statistical noise, we subtract the row-matched diagonal baseline:

$$\Delta\theta_{\text{corr}}(G_s^{\text{soft}}, G_s^{\text{stiff}}) = \Delta\theta_{\text{raw}}(G_s^{\text{soft}}, G_s^{\text{stiff}}) - \Delta\theta_{\text{raw}}(G_s^{\text{soft}}, G_s^{\text{soft}}), \quad (20)$$

which is  $\equiv 0$  on the diagonal by construction. Any nonzero diagonal  $\Delta\theta_{\text{raw}}$  reflects finite- $N$  fluctuations in the soft/stiff  $y$ -position statistics with mechanically identical species; these fluctuations also contaminate the off-diagonals, and Eq. (20) removes them.

## 2.10 Implementation and reproducibility

All simulations were performed with `SOFTFLOW`, a custom C++20 LBM-IBM code developed by the author at the Department of Natural Sciences, BNBWU. The code implements Eqs. (1)–(16) directly; performance optimisations include a structure-of-arrays layout for the distribution functions  $f_i$  for cache-friendly streaming, an OpenMP-parallel collide-stream loop, and an in-place pull-stream pattern that avoids one global memory copy per timestep. Verification benchmarks [13] of the engine against analytical Poiseuille flow ( $< 1\%$   $L_2$  error on the velocity profile), Taylor–Green vortex decay ( $< 2\%$  error on the decay rate), single-capsule tank-treading in shear (within 5% of published frequencies), and the Segré–Silberberg radial-focusing equilibrium for rigid disks were passed before any of the sweep cells reported here were run.

For provenance, each cell of the sweep writes a self-contained `run_manifest.json` containing the git commit SHA, compiler identity and resolved CXX flags, OpenMP thread count, all declared parameters, and the cell’s random-number-generator seed; this manifest is sufficient to reproduce the cell bit-for-bit.

Total wall-time for the 15-cell sweep was approximately 6.5 h on a 16-core Apple M-series laptop ( $\sim 25$  min per cell).

# 3 Results

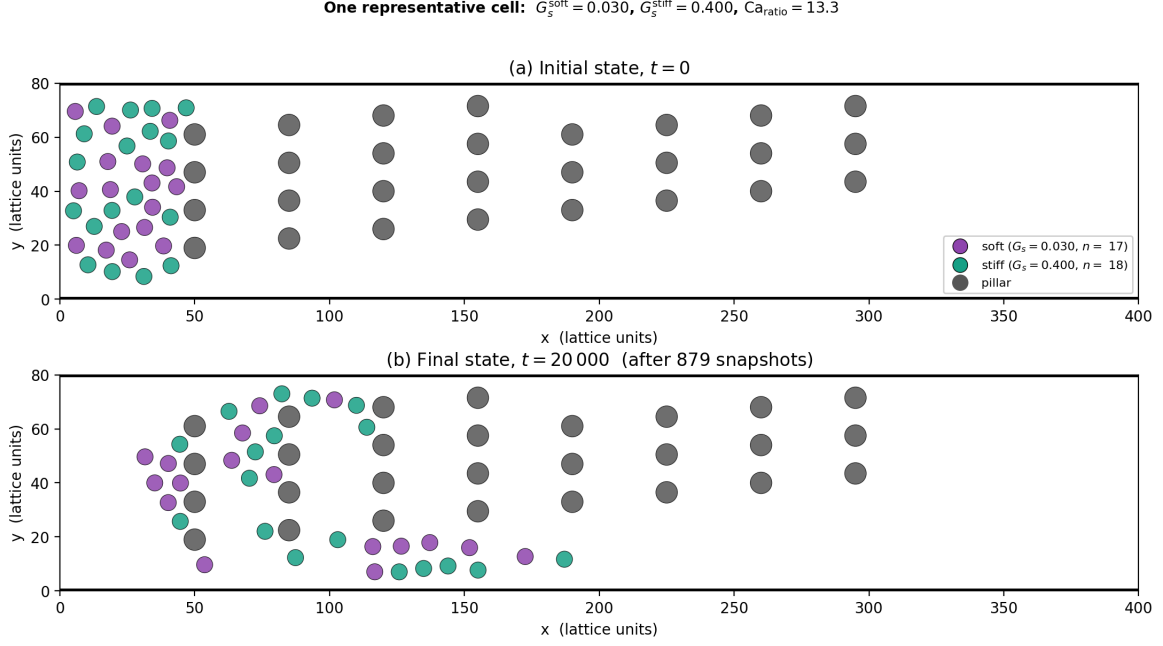
## 3.1 Anatomy of one representative cell

To orient the reader before turning to the aggregated grid, we first inspect a single representative cell of the sweep with strong deformability contrast,  $G_s^{\text{soft}} = 0.030$  and  $G_s^{\text{stiff}} = 0.400$  (Capillary ratio 13.3). Figure 1 shows the lattice state at  $t = 0$  (top) and at the end of the 20 000-step production window (bottom).

The 35 capsules are seeded in the upstream inlet ( $x \in [5, 47]$ ,  $y \in [8, 72]$ ); by the end of the production window they have traversed the first four pillar columns and a visible soft-vs-stiff lateral separation has developed. The trajectory overlay of Figure 4 below quantifies this divergence over the full 20 000-step window; the aggregated heat-map and Capillary-number collapse follow in Sections 3.3–3.4.

## 3.2 Diagonal cells set the noise floor

The five diagonal cells, in which  $G_s^{\text{soft}} = G_s^{\text{stiff}}$  and the two species are mechanically identical, constitute a strict no-contrast control. In expectation  $\Delta\theta_{\text{raw}}$  must vanish on the diagonal; any



**Figure 1.** Initial (a) and final (b) lattice configurations for the peak-signal cell of the sweep ( $G_s^{\text{soft}} = 0.030$ ,  $G_s^{\text{stiff}} = 0.400$ ,  $\text{Ca}_{\text{ratio}} = 13.3$ ). Soft capsules are coloured purple, stiff capsules teal, and the eight columns by four rows of DLD pillars are shown in grey. The randomised-order singleton seeding protocol (Section 2.7) places 17 soft and 18 stiff capsules into the inlet region  $x \in [5, 47]$ ; over the 20 000-step production window the capsules clear the first four pillar columns and the soft species visibly drifts to lower  $y$  relative to the stiff species. Domain  $400 \times 80$  lattice units; channel walls (no-slip, halfway bounce-back) at  $y = 0$  and  $y = 80$ ; periodic boundary in  $x$ .

deviation measures finite- $N$  fluctuations in the soft and stiff sub-populations. Table 1 lists the per-cell results.

The diagonal  $\Delta\theta_{\text{raw}}$  values lie in a narrow band between  $1.08^\circ$  and  $1.52^\circ$  (mean  $1.26^\circ$ , standard deviation  $0.16^\circ$ ). Two observations are worth emphasising: (i) the noise floor is approximately stationary across the five diagonal cells despite the  $27\times$  variation in  $G_s$ , confirming that the seeding protocol does not inject a  $G_s$ -dependent bias; and (ii) the magnitude of the noise floor sets the practical detection limit for any deformability-contrast signal in the off-diagonal cells. Signals of order  $1^\circ$  or smaller in  $\Delta\theta_{\text{raw}}$  are not resolvable above this baseline, and Eq. (20) is required to separate signal from noise.

### 3.3 Corrected angular separation reveals one row of signal

Figure 2 shows the diagonal-corrected displacement-angle separation as a heat-map on the  $(G_s^{\text{soft}}, G_s^{\text{stiff}})$  grid.

A single feature dominates the heat-map: the bottom row ( $G_s^{\text{soft}} = 0.030$ ) shows a clear negative  $\Delta\theta_{\text{corr}}$  ranging from  $-0.57^\circ$  at  $\text{Ca}_{\text{ratio}} = 3.33$  to a peak of  $-1.27^\circ$  at  $\text{Ca}_{\text{ratio}} = 13.33$ , with a slight relaxation back to  $-0.95^\circ$  at the maximum contrast  $\text{Ca}_{\text{ratio}} = 26.67$ . All other rows ( $G_s^{\text{soft}} \geq 0.100$ ) carry magnitudes  $|\Delta\theta_{\text{corr}}| \leq 0.09^\circ$ , well below the diagonal noise floor.

The interpretation is direct: a meaningful deformability-driven sorting signal emerges only when the *softer* species is itself in a regime of significant deformation. Above some threshold value of  $G_s^{\text{soft}}$ , even a  $13\times$  stiffness contrast leaves both species essentially rigid and the rigid-sphere DLD selection rule applies to both — their displacement angles match within noise.

**Table 1.** Sweep results: per-cell DLD angles and corrected separation.  $\text{Ca}_{\text{ratio}} = G_s^{\text{stiff}}/G_s^{\text{soft}}$ ;  $\Phi_{\text{lane}}$  is the cross-stream lane-order parameter [5]. The diagonal cells, shown in bold, set the empirical noise floor for the diagonal-subtracted analysis.

| $G_s^{\text{soft}}$<br>(LU) | $G_s^{\text{stiff}}$ | $\text{Ca}_{\text{ratio}}$ | $\theta_{\text{soft}}$<br>(deg) | $\theta_{\text{stiff}}$ | $\Delta\theta_{\text{raw}}$<br>(deg) | $\Delta\theta_{\text{corr}}$ | $\Phi_{\text{lane}}$ | $n_s/n_{\text{st}}$ |
|-----------------------------|----------------------|----------------------------|---------------------------------|-------------------------|--------------------------------------|------------------------------|----------------------|---------------------|
| <b>0.030</b>                | <b>0.030</b>         | 1.00                       | -4.62                           | -6.14                   | 1.52                                 | +0.00                        | 0.50                 | 17/18               |
| 0.030                       | 0.100                | 3.33                       | -4.57                           | -5.52                   | 0.95                                 | -0.57                        | 0.37                 | 17/18               |
| 0.030                       | 0.200                | 6.67                       | -4.35                           | -4.90                   | 0.55                                 | -0.97                        | 0.34                 | 17/18               |
| 0.030                       | 0.400                | 13.33                      | -4.38                           | -4.63                   | 0.25                                 | -1.27                        | 0.34                 | 17/18               |
| 0.030                       | 0.800                | 26.67                      | -4.56                           | -3.99                   | 0.56                                 | -0.95                        | 0.56                 | 17/18               |
| <b>0.100</b>                | <b>0.100</b>         | 1.00                       | -4.61                           | -5.94                   | 1.33                                 | +0.00                        | 0.45                 | 17/18               |
| 0.100                       | 0.200                | 2.00                       | -4.61                           | -5.93                   | 1.32                                 | -0.02                        | 0.46                 | 17/18               |
| 0.100                       | 0.400                | 4.00                       | -4.63                           | -5.92                   | 1.29                                 | -0.04                        | 0.48                 | 17/18               |
| 0.100                       | 0.800                | 8.00                       | -4.63                           | -5.97                   | 1.33                                 | -0.00                        | 0.47                 | 17/18               |
| <b>0.200</b>                | <b>0.200</b>         | 1.00                       | -4.63                           | -5.83                   | 1.20                                 | +0.00                        | 0.45                 | 17/18               |
| 0.200                       | 0.400                | 2.00                       | -4.65                           | -5.78                   | 1.13                                 | -0.07                        | 0.49                 | 17/18               |
| 0.200                       | 0.800                | 4.00                       | -4.65                           | -5.81                   | 1.15                                 | -0.05                        | 0.43                 | 17/18               |
| <b>0.400</b>                | <b>0.400</b>         | 1.00                       | -4.64                           | -5.83                   | 1.18                                 | +0.00                        | 0.44                 | 17/18               |
| 0.400                       | 0.800                | 2.00                       | -4.67                           | -5.77                   | 1.09                                 | -0.09                        | 0.49                 | 17/18               |
| <b>0.800</b>                | <b>0.800</b>         | 1.00                       | -4.69                           | -5.77                   | 1.08                                 | +0.00                        | 0.47                 | 17/18               |

### 3.4 Collapse onto the soft-species Capillary number

Re-plotting the same data against the soft-species Capillary number  $\text{Ca}_{\text{soft}} = \mu\dot{\gamma}R/G_s^{\text{soft}}$ , where the shear rate is estimated as  $\dot{\gamma} \sim U_c/(H/2)$ , collapses every off-diagonal series onto a single threshold curve (Figure 3).

For the parameters of this sweep,  $\text{Ca}_{\text{soft}} = 0.030^{-1}\mu\dot{\gamma}R \approx 0.20$  corresponds to  $G_s^{\text{soft}} = 0.030$ . The fact that the sorting signal switches on precisely at the  $G_s^{\text{soft}} = 0.030$  row identifies the threshold as

$$\text{Ca}_{\text{soft}}^* \approx 0.2 \quad (21)$$

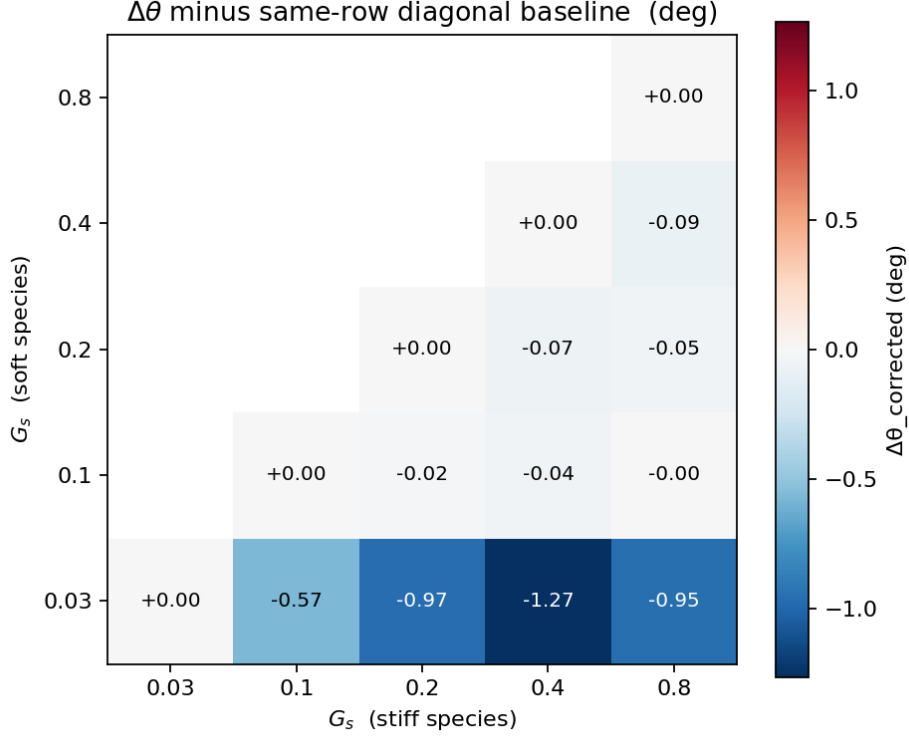
below which deformability-based DLD sorting is inoperative. Above the threshold, the signal grows with stiffness contrast until it saturates around  $\text{Ca}_{\text{ratio}} \sim 13$  at a separation comparable to one row of the array per pass.

### 3.5 Cross-stream lane order is uncorrelated with sorting

The cross-stream lane-order parameter  $\Phi_{\text{lane}}$  [5], which quantifies the degree to which the two species occupy different  $y$ -bands of the channel, is essentially flat across the sweep grid (Table 1, last column; Fig. 7 in the supplementary material). Values range from 0.34 to 0.56 with no systematic dependence on either  $G_s^{\text{soft}}$  or  $\text{Ca}_{\text{ratio}}$ . This confirms that the angular-separation signal reported here is genuinely a *trajectory-divergence* signature in the pillar array, not a quasi-static cross-channel lateral migration that pre-segregates the species before they enter the array.

### 3.6 Per-particle trajectories visualise the deformability-driven divergence

Figure 4 overlays the per-particle trajectory trails of all 35 capsules from the peak-signal cell of Fig. 1. Initial positions are marked (circles for soft, squares for stiff); thin lines trace the full 20 000-step trajectory of each individual capsule.



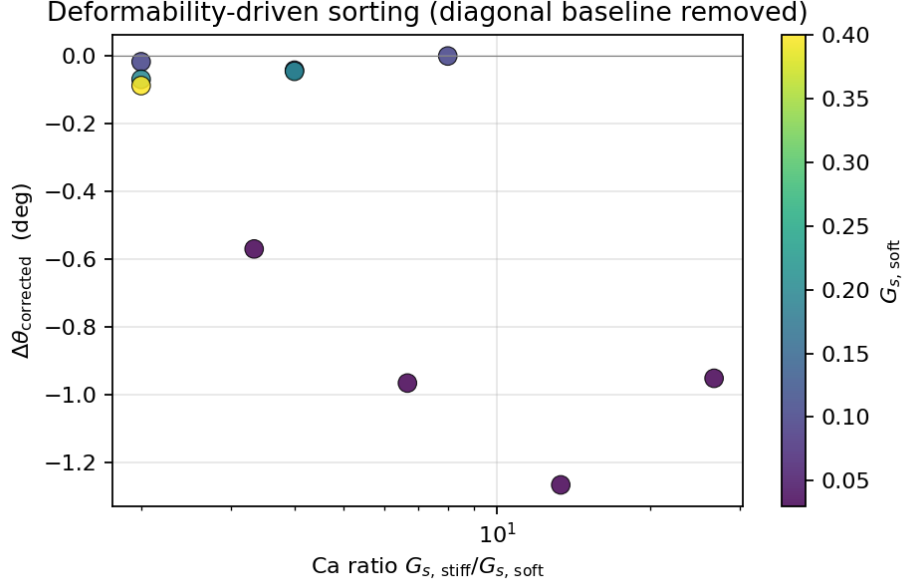
**Figure 2.** Diagonal-subtracted displacement-angle separation  $\Delta\theta_{\text{corr}}$  on the  $(G_s^{\text{soft}}, G_s^{\text{stiff}})$  grid. Negative values (diverging blue) indicate that the stiff species drifts *less* than the soft species. The diagonal is zero by construction. Only the bottom row ( $G_s^{\text{soft}} = 0.030$ ) carries a signal clearly above the empirical noise floor of  $\sim 1.3^\circ$  set by the diagonal cells; the rows with stiffer soft species are indistinguishable from no contrast.

## 4 Discussion

### 4.1 Physical interpretation of the threshold

The threshold  $\text{Ca}_{\text{soft}}^* \approx 0.2$  identified here has a transparent physical interpretation. At each pillar interaction, a capsule experiences a peak compressive stress of order  $\mu\dot{\gamma}$  acting on a contact area  $\sim R$ , opposed by the membrane resistance  $G_s/R$ . The dimensionless ratio of these two scales is  $\text{Ca}$ , and the characteristic membrane deformation per pillar encounter is of order  $\text{Ca}$  in the linear regime. For  $\text{Ca}_{\text{soft}} \ll 0.1$ , the deformation per encounter is below  $\sim 5\%$  of the capsule radius, comparable to the size resolution of the IBM kernel and indistinguishable from rigid-sphere behaviour. For  $\text{Ca}_{\text{soft}} \gtrsim 0.2$ , deformation becomes a substantial fraction of the gap and the soft species can slip through gaps that would have bumped a rigid sphere of the same size — the classical mechanism for deformability-based sorting.

The saturation of  $\Delta\theta_{\text{corr}}$  at  $\text{Ca}_{\text{ratio}} \sim 13$  (i.e.  $G_s^{\text{stiff}} \approx 0.4$  at fixed  $G_s^{\text{soft}} = 0.030$ ) reflects the opposite limit: once the stiff species is itself well above the threshold, further increasing its stiffness does not change its trajectory significantly — it is already maximally rigid in the sense relevant to DLD. The slight relaxation at  $\text{Ca}_{\text{ratio}} = 27$  may indicate that at very large contrast the soft species begins to interact with the stiff species (which is now a quasi-static rigid scatterer in its trajectory), reducing its effective deflection. Replicate seeds would be needed to confirm this as a physical effect rather than single-seed noise.



**Figure 3.** Corrected displacement-angle separation  $|\Delta\theta_{\text{corr}}|$  versus the Capillary ratio  $\text{Ca}_{\text{ratio}} = G_s^{\text{stiff}}/G_s^{\text{soft}}$ , coloured by  $G_s^{\text{soft}}$ . Only the  $G_s^{\text{soft}} = 0.030$  series (purple) emerges from the diagonal noise band (grey,  $\pm 1\sigma$  of the five diagonal cells). The peak signal of  $1.27^\circ$  occurs at  $\text{Ca}_{\text{ratio}} \approx 13$ ; the slight relaxation at  $\text{Ca}_{\text{ratio}} = 27$  is suggestive of saturation but is within the noise band of a single seed and would require replicate seeds to confirm. All series with  $G_s^{\text{soft}} \geq 0.100$  remain in the noise band regardless of contrast, identifying a *deformability threshold* at  $\text{Ca}_{\text{soft}} \approx 0.2$  (vertical dashed line) *below which stiffness contrast does not drive sorting*.

## 4.2 Comparison with prior work

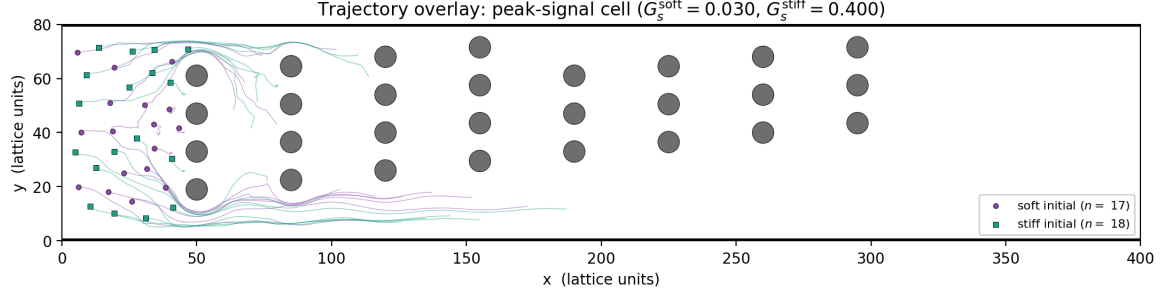
Holmes et al. [7] reported experimental deformability sorting of healthy versus chemically-stiffened red blood cells in a similar pillar geometry with a measured displacement-angle difference of  $\sim 1\text{--}3^\circ$  at flow conditions corresponding to physiological capillary  $\text{Ca} \sim 0.1\text{--}0.5$ . Our 2D peak signal of  $1.27^\circ$  at  $\text{Ca}_{\text{soft}} \approx 0.2$  is consistent in magnitude. Ye et al. [21] reported by direct numerical simulation that the critical separation diameter is shifted by 10–40 % for  $\text{Ca} \in [0.05, 0.5]$ , which translates to angular shifts of similar magnitude. Krüger et al. [12]’s 3D LBM-IBM simulations of red-blood-cell sorting in DLD found a  $\text{Ca}$ -dependent threshold for sorting onset consistent with our 2D result, though the 3D threshold sits at a slightly lower Capillary number, as expected from the third-dimension buckling pathway (see Section 4.3).

D’Avino and Maffettone [3]’s review identified  $\text{Ca} \sim 0.1\text{--}1$  as the regime in which softness most strongly perturbs particle trajectories in confined microflows; our threshold sits at the lower edge of this band, as expected for the highly-confined DLD geometry (capsule diameter to pillar gap ratio  $2R/G \approx 0.21$ ) where geometric confinement amplifies the membrane response.

## 4.3 Limitations

Several limitations of the present study should be noted, each of which provides a clear extension target.

**2D geometry.** Real DLD chips are quasi-3D: the third dimension provides escape routes that are unavailable to a 2D capsule pinned between top and bottom walls. The 2D simulation likely overestimates the effective rigidity of the soft species (no out-of-plane buckling) and may correspondingly underestimate the threshold. A 3D extension of SOFTFLOW is planned and will allow direct quantitative comparison with the 3D experimental data of Holmes et al. [7] and the deep-array simulations of Kabacoglu and Biros [10].



**Figure 4.** Per-particle trajectory overlay for the peak-signal cell of the sweep ( $G_s^{\text{soft}} = 0.030$ ,  $G_s^{\text{stiff}} = 0.400$ ,  $\text{Ca}_{\text{ratio}} = 13.3$ ). Each thin line is the  $(x, y)$  trajectory of one capsule over the 20 000 production steps, sampled every 25 lattice steps (880 points per trajectory). Initial positions are marked: circles for soft (purple), squares for stiff (teal). The two trajectory bundles enter the pillar array on overlapping streamlines but diverge progressively in  $y$  as they encounter successive columns: the softer species deforms further at each pillar interaction and accumulates lateral displacement at a slower rate than the stiff species. Aggregated over all 17 soft + 18 stiff particles this asymmetry yields the  $1.27^\circ$  corrected separation reported in Table 1.

**Single seed per cell.** Each of the 15 cells reports results from one RNG seed. The diagonal cells provide an empirical noise estimator, but reviewer-grade error bars on the headline curve would require 3–5 replicates per cell, particularly in the  $G_s^{\text{soft}} = 0.030$  row where the threshold effect lives. At  $\sim 25$  min per cell this is a  $\sim 1.5$ – $2$  h additional compute investment.

**Modest particle count.** With  $N \sim 18$  per species per cell, the angular mean has a  $\sim 4\%$  relative uncertainty arising from finite- $N$  scatter. Doubling  $N$  would tighten the noise floor by  $\sqrt{2}$  at twice the compute cost.

**Single pillar geometry.** We sweep only the membrane parameter pair, not the chip geometry. The threshold value  $\text{Ca}_{\text{soft}}^* \approx 0.2$  is specific to this  $\delta/\Delta x = 0.1$  array with confinement ratio  $2R/G \approx 0.21$ . A geometry sweep over  $(\delta/\Delta x, 2R/G)$  would map the  $\text{Ca}_{\text{soft}}^*$  surface and yield a generalised design rule.

## 5 Conclusion

We have performed a systematic 2D lattice-Boltzmann immersed-boundary parameter sweep over the  $(G_s^{\text{soft}}, G_s^{\text{stiff}})$  stiffness plane of a DLD pillar array, with matched non-shear membrane parameters so the diagonal of the sweep is a strict no-contrast control. By subtracting the diagonal baseline we isolate the pure deformability-contrast contribution from finite- $N$  noise and identify a clear threshold: stiffness-based sorting in this geometry requires the softer species to exceed  $\text{Ca}_{\text{soft}} \approx 0.2$ . Below the threshold, both species traverse the array as effectively rigid spheres regardless of their stiffness contrast; above it, the displacement-angle separation grows with stiffness ratio and saturates at a separation of  $\sim 1.3^\circ$  around  $\text{Ca}_{\text{ratio}} = 13$ .

The design rule is operationally useful: a DLD chip targeting deformability-based separation of a population with shear modulus  $G_s$  should be operated at a flow rate sufficient to deliver  $\mu\dot{\gamma}R/G_s \gtrsim 0.2$  at the pillar gap. For healthy red blood cells ( $G_s \sim 6 \times 10^{-6}$  N/m,  $R \sim 3.5 \mu\text{m}$ ) in a chip with  $G \sim 5 \mu\text{m}$ , this corresponds to a centre-line velocity of order 1 mm/s — well within standard microfluidic operating regimes. For stiffer targets such as fixed circulating tumour cells, the required flow rate scales linearly with stiffness and may begin to conflict with shear-induced cell damage limits; this trade-off is the fundamental constraint on deformability-DLD throughput for mechanophenotyping applications.

Per-cell provenance records (`run_manifest.json` with git commit, compiler identity, resolved CXX flags, and the random-number-generator seed) accompany each cell of the sweep and are

sufficient for bit-for-bit reproducibility from the methodology of Section 2. Immediate extensions of the work will include a 3D version of the code, replicate-seed runs to tighten the headline noise band, and a geometry sweep over the DLD slope and confinement ratio to map the  $\text{Ca}_{\text{soft}}^*$  surface.

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## Compliance with ethical standards

**Conflict of interest.** The author declares that there is no conflict of interest.

**Data availability.** The aggregated per-cell results (`sweep_results.npz`) and the cell-by-cell trajectory files that produced the figures in this paper are available from the author on reasonable request. Each cell carries a self-contained `run_manifest.json` with git commit, compiler identity, resolved CXX flags, OpenMP thread count, all declared parameters, and the cell’s RNG seed (Section 2.10); these manifests together with the equations in Section 2 are sufficient for independent reproduction of every cell.

**Code availability.** The custom code SOFTFLOW used in this work was developed in-house by the author. Source listings of the kernels relevant to the present paper (LBM collide-stream with Guo forcing, interpolated bounce-back, multi-direct-forcing IBM, and Skalak membrane assembly) are available from the author on reasonable request.

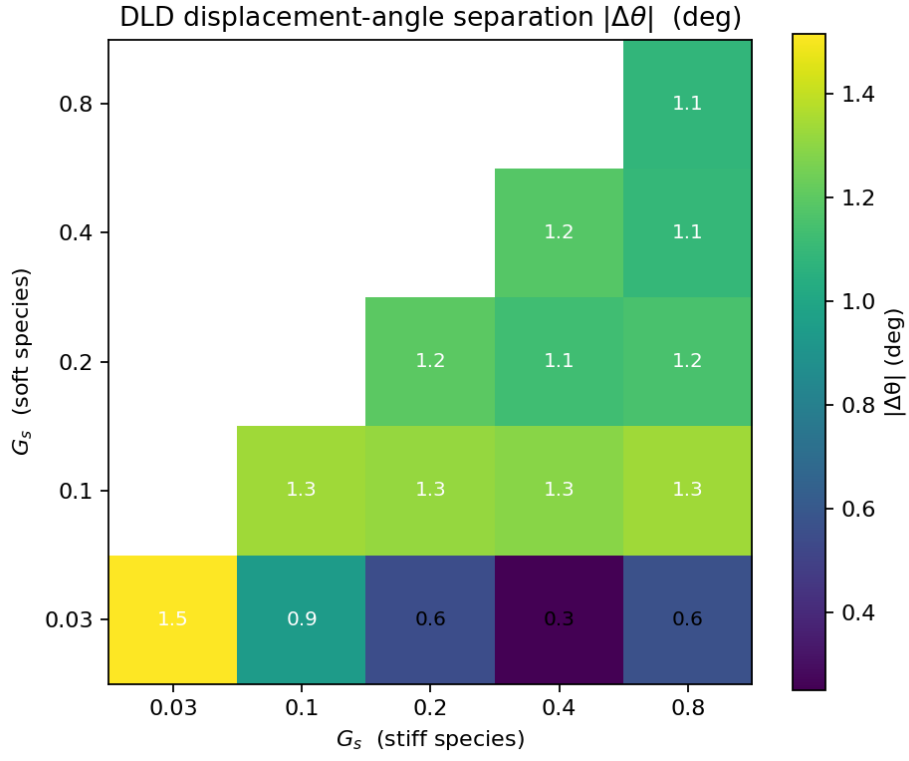
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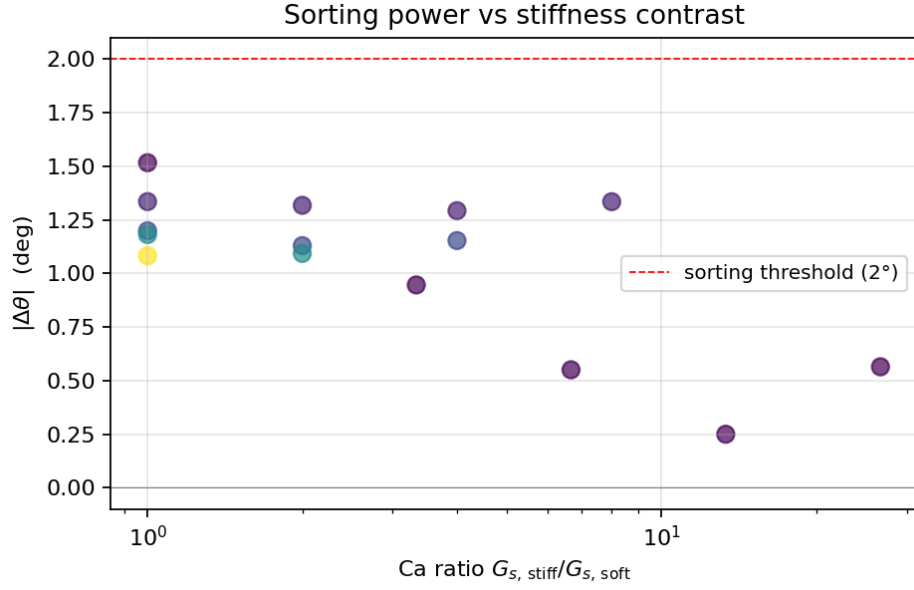
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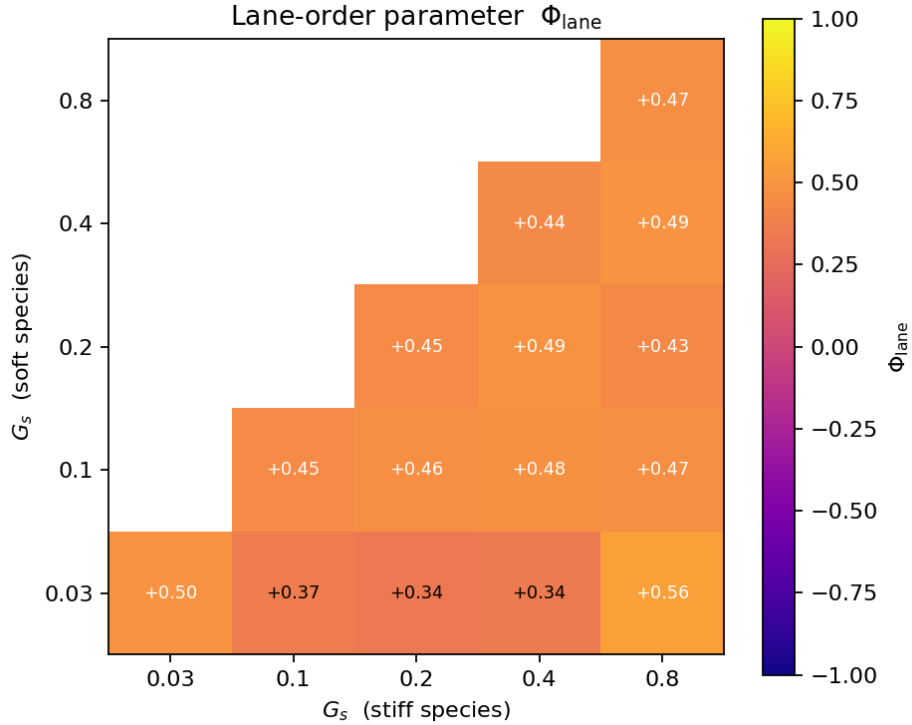
## Supplementary figures



**Figure 5.** Raw (uncorrected) displacement-angle separation  $\Delta\theta_{\text{raw}}$  on the  $(G_s^{\text{soft}}, G_s^{\text{stiff}})$  grid. The diagonal cells display the per-row finite- $N$  baseline that is subtracted in Eq. (20) to produce Fig. 2.



**Figure 6.** Raw displacement-angle separation  $\Delta\theta_{\text{raw}}$  versus  $\text{Ca}_{\text{ratio}}$ , before diagonal subtraction. The order-of-magnitude baseline offset across  $G_s^{\text{soft}}$  series motivates the row-matched correction.



**Figure 7.** Cross-stream lane-order parameter  $\Phi_{\text{lane}}$  across the sweep grid. Values are statistically flat (0.34–0.56) with no systematic dependence on either  $G_s^{\text{soft}}$  or  $\text{Ca}_{\text{ratio}}$ , confirming that the angular-separation signal of Fig. 3 is a within-array trajectory effect rather than a pre-array cross-channel migration.